

ORIGINAL RESEARCH

HEART FAILURE

Left Atrial Volumetric Enlargement in Heart Failure With Reduced Ejection Fraction



Obligatory Consequence or Independent Predictor of Outcome?

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ABSTRACT

BACKGROUND Left atrial (LA) remodeling, characterized by progressive enlargement, is commonly observed in heart failure with reduced ejection fraction and reflects left ventricular (LV) dysfunction and functional mitral regurgitation (FMR). However, its independent prognostic significance remains uncertain.

OBJECTIVES This study aims to investigate the independent association between LA volume index (LAVI) and long-term mortality in patients with heart failure, considering its interaction with FMR severity.

METHODS A cohort of 13,954 patients who received a diagnosis of heart failure stage B or C and a first diagnosis of reduced LV ejection fraction (<50%) at Mayo Clinic was analyzed. Comprehensive echocardiographic quantification of LAVI, LV function, FMR, and hemodynamics was performed. Survival analysis included Kaplan-Meier curves, Cox proportional hazards models, and spline modeling, adjusted for demographic, clinical, and echocardiographic variables.

RESULTS Mean LAVI increased with worsening FMR (35 ± 14 mL/m² in no FMR to 61 ± 23 mL/m² in severe FMR). Higher LAVI was associated with increased mortality (HR: 1.13 per 10 mL/m²; $P < 0.0001$), independent of LV function, FMR, and comorbidities. A mortality risk threshold emerged at LAVI >40 mL/m², with excess mortality increasing steeply beyond this point. Stratification showed significant interactions between LAVI, LV ejection fraction, atrial fibrillation, and pulmonary hypertension. Although LAVI threshold of excess mortality shifted from about 40 mL/m² in no-mild FMR to about 60 mL/m² in severe FMR, the slope of the relationship persisted unaltered, confirming the graded relationship between LAVI and outcome at each FMR grade (P for interaction = 0.35).

CONCLUSIONS LAVI is an independent predictor of mortality in heart failure stage B or C, with excess risk starting at 40 mL/m². The prognostic impact of LAVI persisted with a graded relationship in all FMR severity grades, emphasizing its role in risk stratification. These findings support the integration of LA remodeling assessment into heart failure management to refine prognostic evaluation and therapeutic strategies. (JACC Heart Fail. 2026;14:102946)

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**ABBREVIATIONS
AND ACRONYMS****EF** = ejection fraction**FMR** = functional mitral
regurgitation**HF** = heart failure**HFrEF** = heart failure with
reduced ejection fraction**LA** = left atrial**LAVI** = left atrial volume index**LV** = left ventricular**LVEF** = left ventricular
ejection fraction**MR** = mitral regurgitation

Left atrial (LA) remodeling is characterized by progressive LA enlargement and is strongly associated in the general population with the cardiovascular disease burden.¹ LA remodeling is easily quantifiable in practice by echocardiography as left atrial volume index (LAVI), is similarly applicable in men and women,² and is common in patients with all forms of heart failure (HF).³ In the context of heart failure with reduced ejection fraction (HFrEF), it is considered to reflect the severity and duration of LA pressure elevation.⁴ Thus, LA remodeling has been thought of as useful in overcoming diagnostic limitations of instan-

taneous LA pressure measurements in reflecting the severity of cardiac disease.⁵ Conversely, LA remodeling may just reflect other characteristics, such as left ventricular (LV) systolic dysfunction, LV diastolic dysfunction,⁶ functional mitral regurgitation (FMR),⁷ or atrial arrhythmias.² Hence, LA remodeling due to multiple intertwined determinants remains of uncertain clinical relevance and has not yet been included among components of prominent prognostic HF scores.⁸ Conversely, LA remodeling could be an integrator of cardiac alterations and be incrementally and independently useful in defining the prognosis of patients with HFrEF. Thus, examining the independent association of LA remodeling with outcome is crucial.

In this regard, uncertainty remains about the independent nature of the association between larger LA dimension and reduced survival in HFrEF. Older positive data used surrogates for LA volume (ie, LA linear dimensions or areas⁹⁻¹²) and were small. Even in studies measuring LAVI, those that positively associated with outcome,¹³⁻¹⁶ could not be fully conclusive and were balanced by studies that did not find a significant link between LAVI and outcome.^{17,18}

This uncertainty is heightened because all studies were of modest size and generally lacked extensive covariates/comorbidity characterization to examine the collinearities that require very large data sets to resolve. Furthermore, hemodynamic assessment, including severity of LV dysfunction, FMR, diastolic function, and hemodynamic measures,¹⁹ were generally not fully detailed to resolve whether LAVI is truly independently linked to outcome or is a simple surrogate for the conditions at presentation.

Thus, we gathered a large cohort of patients who had LA volumetric remodeling quantified prospectively at the time of first diagnosis of HFrEF in routine practice, with comprehensive hemodynamics/comorbidity characterization and long-term

follow-up, and examined its association and interactions with long-term survival after the diagnosis of HFrEF.

METHODS

ELIGIBILITY CRITERIA. Patients were eligible if they received a first diagnosis of HF stage B or C²⁰ because of HFrEF, defined by left ventricular ejection fraction (LVEF) <50% measured by transthoracic echocardiography performed at the Mayo Clinic outpatient facility (Rochester, Minnesota, USA) from 2003 to 2011;²¹ were ≥50 years of age at diagnosis; and had comprehensive clinical characterization and echocardiographic characterization during the same visit, particularly with prospective LA volumetric assessment using LAVI measurement. Exclusion criteria were as follows: 1) organic mitral disease (prolapse, flail leaflet, prosthetic valve, or ≥ trivial rheumatic or degenerative mitral valve thickening/calcification); 2) aortic stenosis/regurgitation ≥ moderate (sclerosis not excluded); 3) mitral stenosis ≥ moderate; 4) organic tricuspid valve disease; 5) pericardial, congenital, hypertrophic, or infiltrative (amyloidosis, hemochromatosis, and sarcoidosis) heart disease; or 6) previous valve surgery.

CLINICAL AND ECHOCARDIOGRAPHIC DATA. Patients' medical history and clinical characteristics documented by a physician at our institution were retrieved unaltered from electronic records. Vital signs were measured at the time of echocardiography. Atrial fibrillation presence was detected at the time of electrocardiography. Comorbidities were summarized using the Charlson index.

All echocardiographic examinations were performed in routine clinical practice by multiple trained sonographers (>100) and reviewed by cardiologists (>30) at the Mayo Clinic using diverse commercially available machines.

The imaging uniform protocol included all views from standard windows and systematic measurement of LV dimensions, LVEF, cardiac index, LV filling, and systolic pulmonary pressures guided by the ASE (American Society of Echocardiography) recommendations. All diagnoses of valve diseases or associated conditions in the final report were selected from standard phrases among predefined diagnoses/descriptive statement panels that were uniform for all physicians/sonographers. LAVI was measured using ASE-guided formulas,²² the area-length method, or the modified Simpson biplane method, which in our laboratory were shown to provide similar results.² FMR was integratively

graded as none/trivial, mild, moderate, and severe according to multiparametric assessment.²³ Echocardiographic data (qualitative and quantitative) were retrieved unaltered from original reports via electronic transfer.

FOLLOW-UP. The outcome endpoint was mortality under medical management (all causes). Patients were censored at cardiac surgery/transplantation or defibrillator/ventricular assist device implantation. Procedures performed during follow-up were electronically identified using clinical notes and procedure codes. Occurrence and date of deaths were retrieved using Accurant (LexisNexis), a proprietary resource gathering multiple national sources.

STATISTICAL ANALYSIS. Continuous variables were expressed as mean $\pm$ SD. LAVI was analyzed as a continuous variable in all primary analyses. In addition, LAVI was categorized into 3 groups (<40 , 40 – 60 , and ≥ 60 mL/m²) as this stratification proved more informative in this specific cohort of patients with HFrEF and FMR. Moreover, for completeness, the ASE guideline-recommended categorization is provided in [Supplemental Table 1](#) and [Supplemental Figure 1](#). Group comparisons were performed using analysis of variance with appropriate post hoc multiple comparisons. Trends were tested through the Cochran-Armitage test for trend or regression analysis.

LA volume determinants were selected on the basis of biological plausibility and tested using linear regression models.

Survival rates were estimated using the Kaplan-Meier method and compared using the log-rank test. Cox proportional hazards models were used to adjust the association for clinically relevant variables. Three models were created: unadjusted, limited (adjusted for age and sex), core (adjusted for age, sex, LVEF, deceleration time, FMR, and atrial fibrillation), and comprehensive (adjusted for age, sex, LVEF, deceleration time, FMR, atrial fibrillation, dyspnea, and systolic pulmonary artery pressure). Spline modeling of the risk (univariable analysis based on the hazards model) related to LAVI within the study cohort and stratified by FMR grades was performed.

Subgroup analyses was performed with separate Cox models for each selected subgroup. *P* values for interaction were derived from models including multiplicative interaction terms between LAVI and each subgroup variable. Analyses were performed with JMP version 18, R (R Core Team), and SAS (SAS Institute Inc). *P* < 0.05 was considered significant.

RESULTS

The cohort included 13,954 consecutive patients with a first diagnosis of HFrEF and prospective LAVI quantitation.

GENERAL CHARACTERISTICS OF THE COHORT. The general characteristics of the enrolled patients are presented in [Table 1](#). It shows the usual HFrEF characteristics, with age 68 ± 14 years, male predominance, frequent diabetes (26%), systemic hypertension (57%), and dyslipidemia (50%). Patients were in HF stage B in 48% and stage C in 52%. Regarding cardiac medical history, previous coronary artery disease was mentioned in 64% and previous myocardial infarction in 32%, as expected in HFrEF. Although renal function was on average in the normal range, comorbidities were frequent, with the average Charlson index around 3. The medical treatment of these patients was generally intensive, involving combinations of several therapeutic classes ([Table 1](#)). Use of mineralocorticoid receptor antagonists was limited in this cohort.

According to echocardiographic assessment ([Table 2](#)), the LV was profoundly abnormal, with markedly enlarged size and LVEF $36\% \pm 10\%$. Other hemodynamic abnormalities were also considerable with altered diastolic LV filling with $E/e' 17 \pm 9$, frequent FMR of moderate or severe grade in 30%, and an estimated systolic pulmonary pressure of 41 ± 14 mm Hg. Overall, the LA was notably dilated, with volume 86 ± 36 mL and LAVI 44 ± 18 mL/m².

Stratification by LAVI ranges showed that LAVI was <40 mL/m² in 6,872 patients (49%), 40 – 59 mL/m² in 5,018 (36%), and ≥ 60 mL/m² in 2,064 (15%). Stratification by LA enlargement showed profound clinical differences ([Table 1](#)); as LA enlargement increased, patients were older, were more often male, and had more cardiovascular risk factors (hypertension, dyslipidemia, and diabetes). Although chronic pulmonary disease was similarly distributed, comorbidities were more frequent with LA enlargement, with a higher Charlson index. Although blood pressure and heart rate were similar, HF was also more severe with LA enlargement, with more dyspnea, edema, atrial fibrillation, altered renal function, and higher MAGGIC (Meta-Analysis Global Group in Chronic) risk score, despite more intensive medical therapy. For all echocardiographic characteristics ([Table 2](#)), there was a significant trend for more severe alterations with increasing LA enlargement, including LV size, systolic function, output, diastolic filling, FMR, and hemodynamics with higher systolic pulmonary pressure. Thus, it is amply

TABLE 1 Clinical Characteristics of the Study Cohort

	Overall (N = 13,954)	LAVI <40 mL/m ² (n = 6,872)	LAVI 40-59 mL/m ² (n = 5,018)	LAVI ≥60 mL/m ² (n = 2,064)	P Value for Trend
Age, y	68 ± 14	65 ± 15	71 ± 13	73 ± 13	0.0001
Male	69	65	72	77	0.0001
BSA, m ²	1.98 ± 0.26	1.98 ± 0.26	1.98 ± 0.26	1.97 ± 0.26	0.07
BMI, kg/m ²	28.8 ± 6.3	28.9 ± 6.5	29.0 ± 6.2	28.1 ± 5.8	0.0004
Heart rate, beats/min	75.3 ± 17.7	75.0 ± 17.5	75.5 ± 18.1	75.5 ± 17.3	0.25
SBP, mm Hg	122 ± 21	121 ± 21	123 ± 21	121 ± 22	0.15
DBP, mm Hg	69.8 ± 13.1	69.8 ± 12.8	69.9 ± 13.4	69.3 ± 13.4	0.28
Diabetes	3,568 (26)	1,586 (24)	1,468 (30)	514 (25)	0.0001
HTN	8,015 (57)	3,675 (53)	3,123 (62)	1,217 (59)	0.0001
Dyslipidemia	6,944 (50)	3,432 (50)	2,575 (51)	937 (45)	0.02
History of CAD	8,991 (64)	4,245 (62)	3,389 (68)	1,357 (66)	0.0001
History of MI	4,443 (32)	2,318 (34)	1,630 (32)	495 (24)	0.0001
History of PCI/CABG	5,503 (39)	2,583 (38)	2,112 (42)	809 (39)	0.003
COPD	2,110 (15)	1,067 (16)	738 (15)	305 (15)	0.42
Charlson index	2.97 ± 2.56	2.77 ± 2.54	3.20 ± 2.62	3.07 ± 2.44	0.0001
Atrial fibrillation	2,707 (20)	583 (9)	1,212 (25)	912 (46)	0.0001
Dyspnea	7,214 (52)	3,142 (46)	2,791 (56)	1,281 (62)	0.0001
Angina	3,533 (25)	1,960 (29)	1,189 (24)	384 (19)	0.0001
Edema	4,019 (29)	1,512 (22)	1,691 (34)	816 (40)	0.0001
Jugular venous distention	1,045 (7)	302 (4)	456 (9)	287 (14)	0.0001
MAGGIC risk score	18.9 ± 7.5	16.8 ± 7.2	20.9 ± 7.1	23.0 ± 7.1	0.0001
Laboratory					
Hemoglobin, g/dL	12.9 ± 2.0	13.1 ± 2.0	12.7 ± 2.0	12.7 ± 2.0	0.0001
Creatinine, mg/dL	1.40 ± 1.01	1.28 ± 0.89	1.51 ± 1.15	1.52 ± 0.96	0.0001
GFR, mL/min/m ²	76.2 ± 41.3	79.9 ± 41.3	67.7 ± 37.4	61.8 ± 33.8	0.0001
Medications					
ACEI/ARB	10,837 (78)	5,122 (75)	4,021 (80)	1,694 (82)	0.0001
Beta-blockers	11,454 (82)	5,343 (78)	4,316 (86)	1,795 (87)	0.0001
Diuretic agents	9,207 (66)	3,771 (55)	3,722 (74)	1,714 (83)	0.0001
MRA	2,201 (16)	878 (12)	847 (16)	476 (23)	0.0001

Values are mean ± SD or n (%), unless otherwise indicated.

ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin-receptor blocker; BMI = body mass index; BSA = body surface area; CABG = coronary artery bypass graft; CAD = coronary artery disease; COPD = chronic obstructive pulmonary disease; DBP = diastolic blood pressure; GFR = glomerular filtration rate; HTN = hypertension; LAVI = left atrial volume index; MAGGIC = Meta-Analysis Global Group in Chronic; MI = myocardial infarction; MRA = mineralocorticoid receptor antagonist; PCI = percutaneous coronary intervention; SBP = systolic blood pressure.

obvious that LA enlargement is not an isolated phenomenon and is part of the more severe presentation of HFrEF.

LAVI-ASSOCIATED CLINICAL AND ECHOCARDIOGRAPHIC CHARACTERISTICS. The clinical and echocardiographic characteristics were entered into a multivariable analysis of the determinants of LA remodeling measured by LAVI. Although candidate variables such as history of coronary disease, history of hypertension, or systolic blood pressure were considered, they did not reach significance in multivariable analysis. The determinants of higher LAVI are presented in [Table 3](#) in order of importance, with the largest impact related to atrial fibrillation. The LV factors are focused more on diastolic physiological determinants, higher LV mass index, higher E/e', and shorter mitral deceleration time, all suggesting that

higher LV filling pressure are linked to larger LAVI whereas lower LVEF is much less strong. The degree of FMR is a key determinant of LA enlargement (LAVI was 35 ± 14 mL/m² in patients with no FMR, 43 ± 15 mL/m² in those with mild FMR, 50 ± 17 mL/m² in those with moderate FMR, and 61 ± 23 mL/m² in those with severe FMR), whereas older age and male sex play a more limited role. Finally, reduced renal function is associated with larger LAVI, probably through volume interactions. Of note, the overall R² of the model was modest (R² = 0.34), suggesting that much LA remodeling remains individually variable and with potential for independent impact on outcome. [Table 3](#) presents the ORs for each determinant linked to more pronounced LAVI enlargement, showing increasing ORs and higher areas under the receiver-operating characteristic curve for severe compared with moderate LAVI enlargement.

TABLE 2 Echocardiographic Characteristics of the Study Cohort

	Overall (N = 13,954)	LAVI <40 mL/m ² (n = 6,872)	LAVI 40-59 mL/m ² (n = 5,018)	LAVI ≥60 mL/m ² (n = 2,064)	P Value for Trend
LV wall thickness, mm	11.0 ± 2.2	10.7 ± 2.1	11.1 ± 2.1	11.3 ± 2.4	0.0001
LVEDD, mm	56.8 ± 8.6	54.5 ± 7.5	57.8 ± 8.7	60.6 ± 9.7	0.0001
LVEDDi, mm/m ²	28.92 ± 4.86	27.9 ± 4.4	29.4 ± 4.8	31.2 ± 5.6	0.0001
LVESD, mm	45.9 ± 9.5	43.4 ± 8.0	47.3 ± 9.7	50.6 ± 10.8	0.0001
LVESDi, mm/m ²	23.48 ± 5.15	22.2 ± 4.5	24.1 ± 5.1	26.0 ± 6.0	0.0001
LVMI, g/m ²	126 ± 37	115 ± 33	132 ± 36	146 ± 41	0.0001
EF, %	35.7 ± 9.8	37.8 ± 8.9	34.4 ± 10.0	32.1 ± 10.6	0.0001
EF <35%	5,651 (41)	2,209 (32)	2,281 (45)	1,161 (56)	0.0001
WMSI	2.01 ± 0.44	1.94 ± 0.42	2.05 ± 0.45	2.12 ± 0.45	0.0001
SV index	38.28 ± 10.37	39.0 ± 9.6	38.0 ± 10.9	36.8 ± 11.4	0.0001
E, m/s	0.81 ± 0.28	0.72 ± 0.26	0.87 ± 0.28	0.93 ± 0.28	0.0001
E/A	1.26 ± 0.88	1.05 ± 0.69	1.46 ± 0.98	1.86 ± 1.06	0.0001
DTE, ms	193.7 ± 62.03	208 ± 62	186 ± 62	168 ± 52	0.0001
E/e'	16.7 ± 9.1	13.8 ± 7.2	18.9 ± 9.6	21.0 ± 10.5	0.0001
LA volume, mL	85.9 ± 36.5	60.3 ± 14.9	96.5 ± 16.9	145.8 ± 39.3	0.0001
LAVI, mL/m ²	43.5 ± 17.6	30.5 ± 6.3	48.6 ± 5.7	74.2 ± 17.9	0.0001
SPAP, mm Hg	41 ± 14	36 ± 12	43 ± 14	48 ± 15	0.0001
≥Moderate FMR	5,167 (30)	1,028 (15)	1,927 (38)	1,212 (59)	0.0001
RV dysfunction	2,111 (15)	613 (9)	923 (18)	575 (28)	0.0001
PHTN	2,867 (24)	699 (13)	1,345 (29)	823 (41)	0.0001

Values are mean ± SD or n (%), unless otherwise indicated.
DTE = E wave deceleration time; EF = ejection fraction; FMR = functional mitral regurgitation; LA = left atrial; LV = left ventricular; LVEDD = left ventricular end-diastolic diameter; LVEDDi = left ventricular end-diastolic diameter index; LVESD = left ventricular end-systolic diameter; LVESDi = left ventricular end-systolic diameter index; LVMI = left ventricular mass index; PHTN = pulmonary hypertension; RV = right ventricular; SPAP = systolic pulmonary artery pressure; SV = stroke volume; WMSI = wall motion score index; other abbreviation as in [Table 1](#).

Higher LAVI is also associated with worse clinical and echocardiographic presentation. [Table 4](#) displays the clinical association of higher LAVI with more frequent presentation with dyspnea, edema, and jugular venous distention, independently of comprehensive adjustment. It also displays the Doppler echocardiographic association of higher LAVI with more frequent presentation with pulmonary hypertension, right ventricular dysfunction, and ≥ moderate tricuspid regurgitation, independently of

comprehensive adjustment. Furthermore, higher LAVI was associated with higher odds of these markers of advanced presentations.

LAVI AND SURVIVAL AFTER THE DIAGNOSIS OF HFREF. Survival under medical management throughout follow-up was 59% ± 1% at 5 years and 36% ± 1% at 10 years. Spline curve analysis is displayed in [Figure 1](#), showing a very steep curve relationship between LAVI increments and increasing

TABLE 3 Multivariable Determinants of LAVI

	LAVI as a Continuous Variable		LAVI 40-60 mL/m ²		LAVI ≥60 mL/m ²	
	Mean Effect	P Value	OR (95% CI)	P Value	OR (95% CI)	P Value
Atrial fibrillation	+12.7 mL/m ²	<0.0001	3.35 (2.96-3.79)	<0.0001	9.76 (8.29-11.5)	<0.0001
LVMI per 10 g/m ²	+1.2 mL/m ²	<0.0001	1.13 (1.11-1.14)	<0.0001	1.22 (1.20-1.25)	<0.0001
≥Moderate FMR	+4.1 mL/m ²	<0.0001	2.46 (2.22-2.72)	<0.0001	4.83 (4.18-5.59)	<0.0001
E/e' >14	+4.0 mL/m ²	<0.0001	1.95 (1.79-2.14)	<0.0001	2.21 (1.91-2.55)	<0.0001
Age per 10 y increment	+1.4 mL/m ²	<0.0001	1.25 (1.20-1.29)	<0.0001	1.28 (1.21-1.36)	<0.0001
Male	+1.4 mL/m ²	<0.0001	1.59 (1.44-1.75)	<0.0001	2.06 (1.76-2.41)	<0.0001
DTE ≤140 ms	+3.2 mL/m ²	<0.0001	1.68 (1.48-1.92)	<0.0001	2.11 (1.76-2.53)	<0.0001
GFR <60 mL/min/m ²	+1.1 mL/m ²	0.0002	1.12 (1.02-1.24)	0.02	1.28 (1.10-1.49)	0.002
LVEF per 10% decrement	+0.31 mL/m ²	0.035	1.08 (1.02-1.13)	0.003	1.16 (1.07-1.25)	<0.0001
	R ² = 0.34		AUC = 0.77		AUC = 0.90	

AUC = area under the receiver-operating characteristic curve; LVEF = left ventricular ejection fraction; other abbreviations as in [Tables 1 and 2](#).

TABLE 4 LAVI-Associated Presentation

	LAVI 40-60 mL/m ²		LAVI ≥60 mL/m ²	
	OR (95% CI)	P Value	OR (95% CI)	P Value
Dyspnea at diagnosis	1.19 (1.09-1.30)	0.0002	1.30 (1.14-1.49)	<0.0001
Peripheral edema	1.41 (1.27-1.56)	<0.0001	1.57 (1.37-1.80)	<0.0001
Jugular venous distention	1.43 (1.19-1.72)	0.0001	1.73 (1.37-2.18)	<0.0001
SPAP ≥50 mm Hg	1.85 (1.63-2.10)	<0.0001	2.76 (2.35-3.24)	<0.0001
RV dysfunction	1.55 (1.35-1.78)	<0.0001	2.05 (1.71-2.45)	<0.0001
Moderate-severe TR	1.87 (1.67-2.11)	<0.0001	3.11 (2.67-3.62)	<0.0001

Adjusted for age, sex, EF, E/e', LVMI, GFR, AF, and FMR.

AF = atrial fibrillation; TR = tricuspid regurgitation; other abbreviations as in Tables 2 and 3.

mortality, with a very narrow 95% CI. Furthermore, excess mortality (spline curve crossing an HR of 1) occurs around LAVI 40 mL/m², with no excess mortality in the cohort with low LAVI (<40 mL/m²), notable excess mortality for LAVI 40 to 60 mL/m², and, around 60 mL/m², the curve crosses an HR of 1.5, indicating more than a 50% increase in mortality, justifying the LAVI thresholds selected (as well as previous values reported in the published reports).

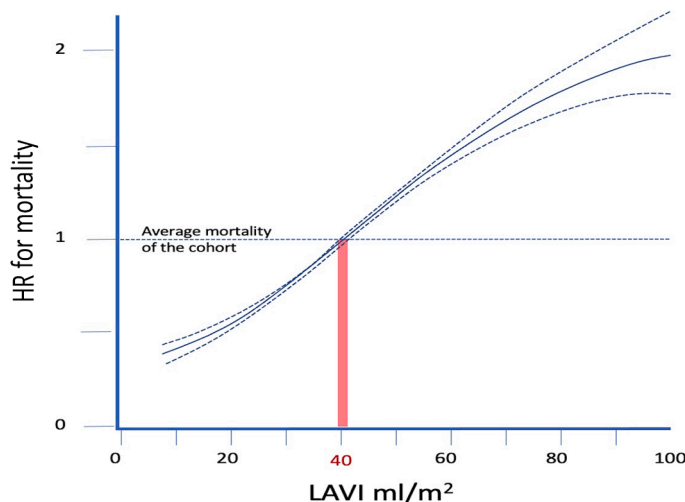
Stratified by these LAVI values, 10-year survival was considerably different (Figure 2): only 17% ± 1% in patients with severe LAVI enlargement (≥60 mL/m²) vs 27% ± 1% in patients with moderate LAVI

enlargement (40-60 mL/m²) vs 46% ± 1% in patients with the smallest LAVI (<40 mL/m²) ($P < 0.0001$).

In Cox proportional hazards analysis, higher LAVI, as a continuous variable, was associated with higher long-term mortality (univariable HR: 1.13 per 10 mL/m² [95% CI: 1.12-1.14 per 10 mL/m²]). Adjusted for core variables (age, sex, dyspnea, and comorbidity index), the HR related to LAVI was slightly reduced but remained highly significant with a narrow 95% CI (Table 5). Even after extensive additional adjustment (core variables + atrial fibrillation, LV mass index, ejection fraction [EF], E/e', deceleration time, moderate/severe FMR, and glomerular filtration rate), the impact of LAVI remained highly significant (adjusted HR: 1.044 per 10 mL/m² [95% CI: 1.027-1.061 per 10 mL/m²]; $P < 0.0001$). Because of the considerable power of our cohort, we were able to conduct models with additional adjustments. With adjustment for the extensive model plus additional adjustment for right-sided variables (pulmonary pressure, right ventricular dysfunction, and tricuspid regurgitation), LAVI remained highly significant (adjusted HR: 1.029 per 10 mL/m² [95% CI: 1.011-1.048 per 10 mL/m²]; $P < 0.0001$). Similarly, additional adjustment for the MAGGIC score (adjusted HR: 1.039 per 10 mL/m² [95% CI: 1.017-1.062 per 10 mL/m²]; $P < 0.0001$) or for congestion signs did not affect LAVI significance (adjusted HR: 1.04 per 10 mL/m² [95% CI: 1.01-1.06 per 10 mL/m²]; $P < 0.0001$). Furthermore, models examined the independent impact of LAVI categorical grades (40-60 mL/m² and ≥60 mL/m² vs <40 mL/m²), showing a highly significant impact for LAVI ≥60 mL/m² across any adjustment (Table 5) and also for LAVI 40 to 60 mL/m² with extensive adjustment, and borderline significance for additional adjustments (Table 5).

Stratification for the main echocardiographically and clinically defined subsets is reported in Figure 3. In all these subsets, higher LAVI was associated with significant excess mortality, demonstrating that the impact of LAVI on outcome is not only independent but also valid across all subsets of patients. Of note, there was a significant interaction between cardiac rhythm and the prognostic value of LAVI, with a lower HR in patients with atrial fibrillation (HR 1.03 per 10 mL/m² [95% CI: 1.01-1.06 per 10 mL/m²]; $P < 0.0001$) vs those in sinus rhythm (HR 1.10 per 10 mL/m² [95% CI: 1.08-1.11 per 10 mL/m²]; $P < 0.0001$), without affecting the significance of LAVI in any of the groups.

The following section analyzes in detail the impact of LAVI in patients with different FMR grades, which directly affects the value of LAVI.

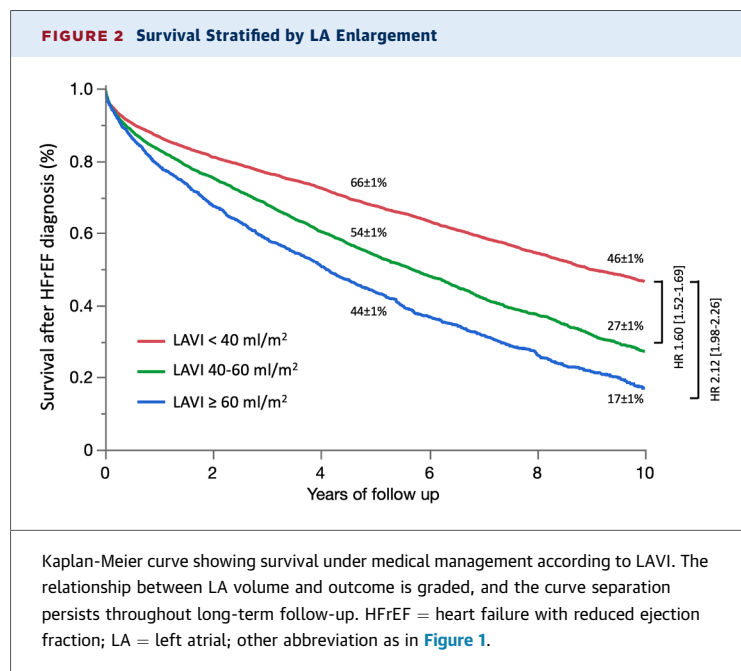
FIGURE 1 Spline Curves of Mortality Risk According to LAVI

The line at HR = 1 represents the average cohort mortality, with values >1 indicating excess mortality. LAVI values shown on the x-axis. Excess mortality under medical management appears at LAVI approximately ≥40 mL/m² and increases rapidly with further LAVI dilation. The curve is based on univariable analysis. LAVI = left atrial volume index.

LAVI OUTCOME ACCORDING TO FMR GRADE. Overall, FMR was present in 67% of patients ($n = 9,335$) and graded as mild in 37%, moderate in 23%, and severe in 7%. The average LAVI increased progressively across the FMR grades: from 35 ± 13 mL/m² in patients with no FMR, 43 ± 15 mL/m² in those with mild FMR, 50 ± 17 mL/m² in those with moderate FMR, and 61 ± 23 mL/m² in those with severe FMR (Figure 4), demonstrating increasing LAVI categorical enlargement but with wide individual variation. In these patients with increasing FMR and LAVI, the yearly mortality rate progressively increased with FMR: 8.1% in no FMR, 10.8% in mild FMR, 14.9% in moderate FMR, and 24.7% in severe FMR. This complex interaction between FMR and LAVI outcome is analyzed in the Central Illustration, showing spline curve analysis for LAVI-associated mortality risk of outcome by FMR strata. Although FMR, LAVI, and risk increase simultaneously, within each FMR stratum the slope of the spline curves is quite similar (P for interaction = 0.35), demonstrating that irrespective of FMR grade and the associated mortality risk, any increment in LAVI is associated with a similar increment in risk, underscoring the complementarity of FMR and LAVI assessment in estimating the risk in patients who received a diagnosis of HFrEF.

DISCUSSION

This large extensively characterized cohort of patients who received a diagnosis of HFrEF in routine practice provides key insights into the complex determinants, context, and prognostic significance of LA remodeling measured by LAVI. The considerable power of the study allows the following first major conclusions: First, LA remodeling determinants are multifactorial, dominated by physiological factors, atrial fibrillation, markers of elevated LV filling pressures (LV mass index and “restrictive filling”), FMR severity, and, to a lesser degree, by poor renal function as a volume marker, with aging having a significant but less important role. However, despite these multiple determinants demonstrated for the first time by our study, an important result is that there is notable individual variation in the degree of LA remodeling in HFrEF. Similarly, LA remodeling is not an isolated phenomenon but is observed along with frequent concomitant clinical signs of HF and right-sided hemodynamic alterations. However, the most important finding is that LAVI is a strong prognostic marker, with significant excess mortality associated with increments of LA remodeling, independently of all associated factors and prognostic



markers, and even of the established MAGGIC score, which emphasizes that individual variation in LA response to LV dysfunction is meaningful and incrementally linked to the outcome in patients with HFrEF. Furthermore, although the interaction between FMR and LA remodeling is difficult to unravel because of their collinearity, the large size and extensive characterization of the present cohort allows us to demonstrate that LAVI retains its prognostic power in each FMR grade, with the slope of the spline curves unchanged, meaning that each LAVI increment yields similar mortality increments irrespective of FMR severity. Therefore, LAVI measurement is a crucial tool for defining high-risk HFrEF in routine practice and should be widely implemented and included in comprehensive prognostic scores for these patients.

COMPLEX DETERMINANTS OF LA REMODELING IN HF. The physiological function of the LA is complex and poorly understood in health and disease, and the role of LA pressure⁴ and its variations is considered essential.²⁴ In patients with HF, the LA is remodeled, more so in patients with HFrEF than in those with preserved EF, with a differential physiology involving a stiff LA in patients with HF and preserved EF.²⁵ In HFrEF, LV compliance decreases because of LV fibrotic remodeling and diastolic dysfunction, with LA pressure overload leading to progressive LA dilation,^{6,26} and the differential potential of LA remodeling was further suggested by the trial of atrial shunting in HF, which suggested an exclusive

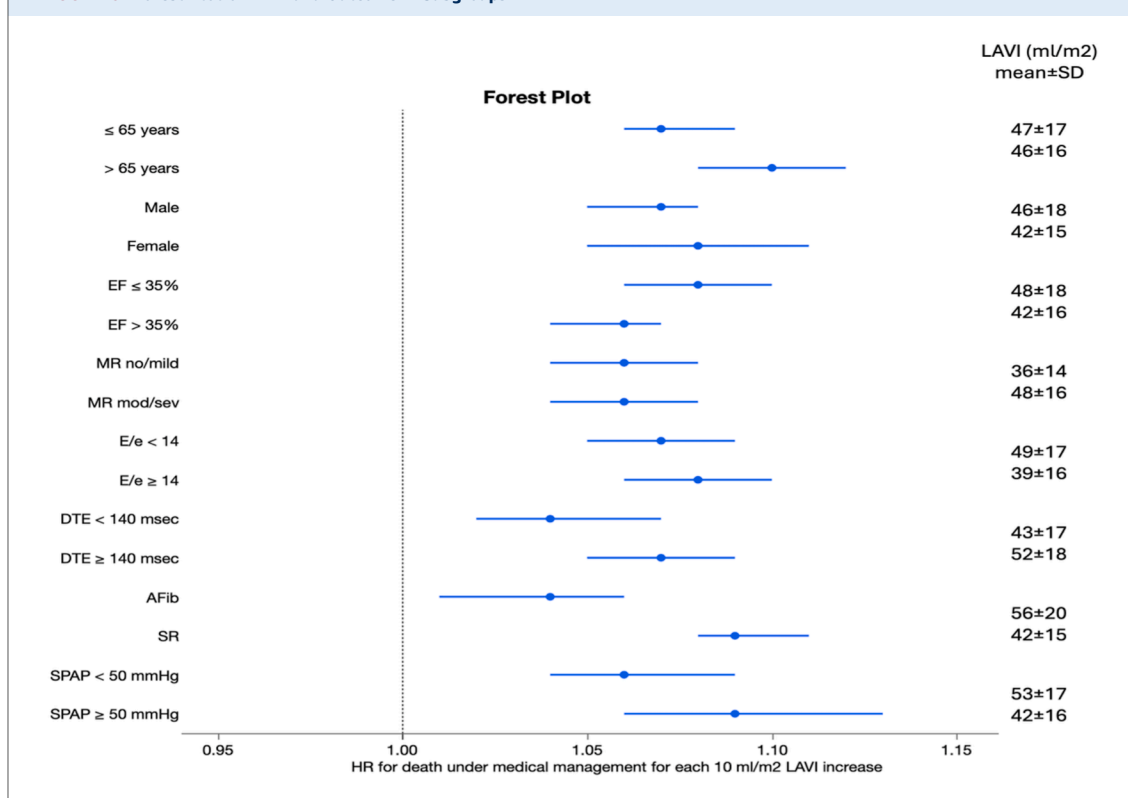
TABLE 5 Association of LAVI With Long-Term Survival in HFrEF

	LAVI continuous per 10 mL/m ²	P Value	LAVI 40-60 mL/m ²	P Value	LAVI ≥60 mL/m ²	P Value
Unadjusted	1.13 (1.12-1.14)	<0.0001	1.60 (1.52-1.69)	<0.0001	2.12 (1.98-2.26)	<0.0001
Core variable adjustment ^a	1.083 (1.071-1.097)	<0.0001	1.21 (1.15-1.28)	<0.0001	1.52 (1.42-1.62)	<0.0001
Extensive adjustment ^b	1.044 (1.027-1.061)	<0.0001	1.08 (1.02-1.15)	0.012	1.22 (1.13-1.33)	<0.0001
Extensive + SPAP, RV dysfunction, TR	1.029 (1.011-1.048)	0.001	1.05 (0.99-1.24)	0.11	1.14 (1.05-1.25)	0.003
Extensive + MAGGIC score	1.039 (1.017-1.062)	0.0006	1.08 (0.99-1.17)	0.08	1.25 (1.11-1.40)	0.003

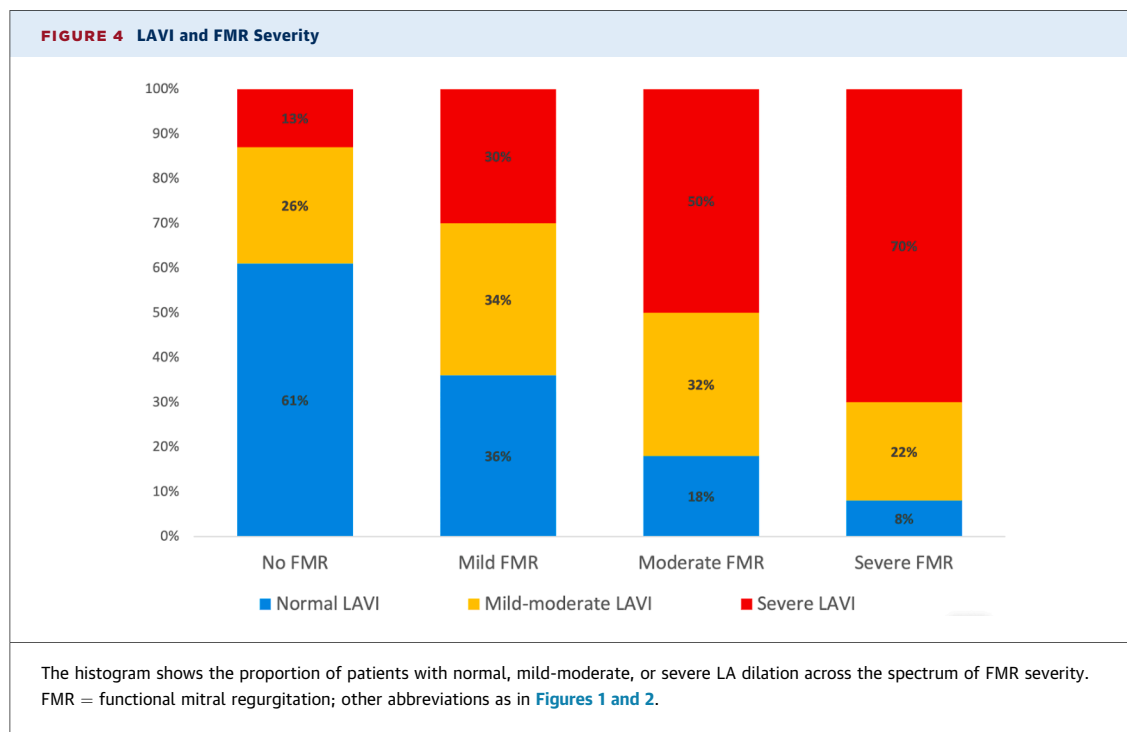
^aAdjustment for age, sex, dyspnea, and comorbidity index. ^bAdjustment for core variables + AF, LVMI, EF, E/e', DT, moderate/severe FMR, and GFR.
HFrEF = heart failure with reduced ejection fraction; other abbreviations as in [Tables 1-3](#).

benefit in patients with HFrEF rather than in those with preserved EF.²⁷ LA remodeling indeed reflects other characteristics, such as LV systolic dysfunction severity, but most strongly in our study, LV hypertrophy,^{28,29} LV diastolic dysfunction,⁶ FMR,^{7,13} and arrhythmias,² such as AF.³⁰ Thus, previous data and particularly our study, with its considerable power and detailed measurements, show that LA remodeling is the integrator of multiple intertwined

physiological determinants, as well as, less importantly, of age and sex. However, the uncertainty that remained before our study is whether LAVI is just a surrogate for these multiple determinants when they entered in a single model or whether it is affected by prominent individual variability in its response to these determinants. This doubt regarding the clinical relevance of LAVI and its significant departure from a simple surrogate prevented its inclusion among

FIGURE 3 Forest Plot of LAVI and Outcome in Subgroups

Subgroup analyses showing HRs for mortality per 10 mL/m² increase in LAVI within each subgroup. Estimates are derived from separate Cox models for each subgroup. AFib = atrial fibrillation; DTE = E wave deceleration time; EF = ejection fraction; mod = moderate; MR = mitral regurgitation; sev = severe; SPAP = systolic pulmonary artery pressure; SR = sinus rhythm; other abbreviation as in [Figure 1](#).

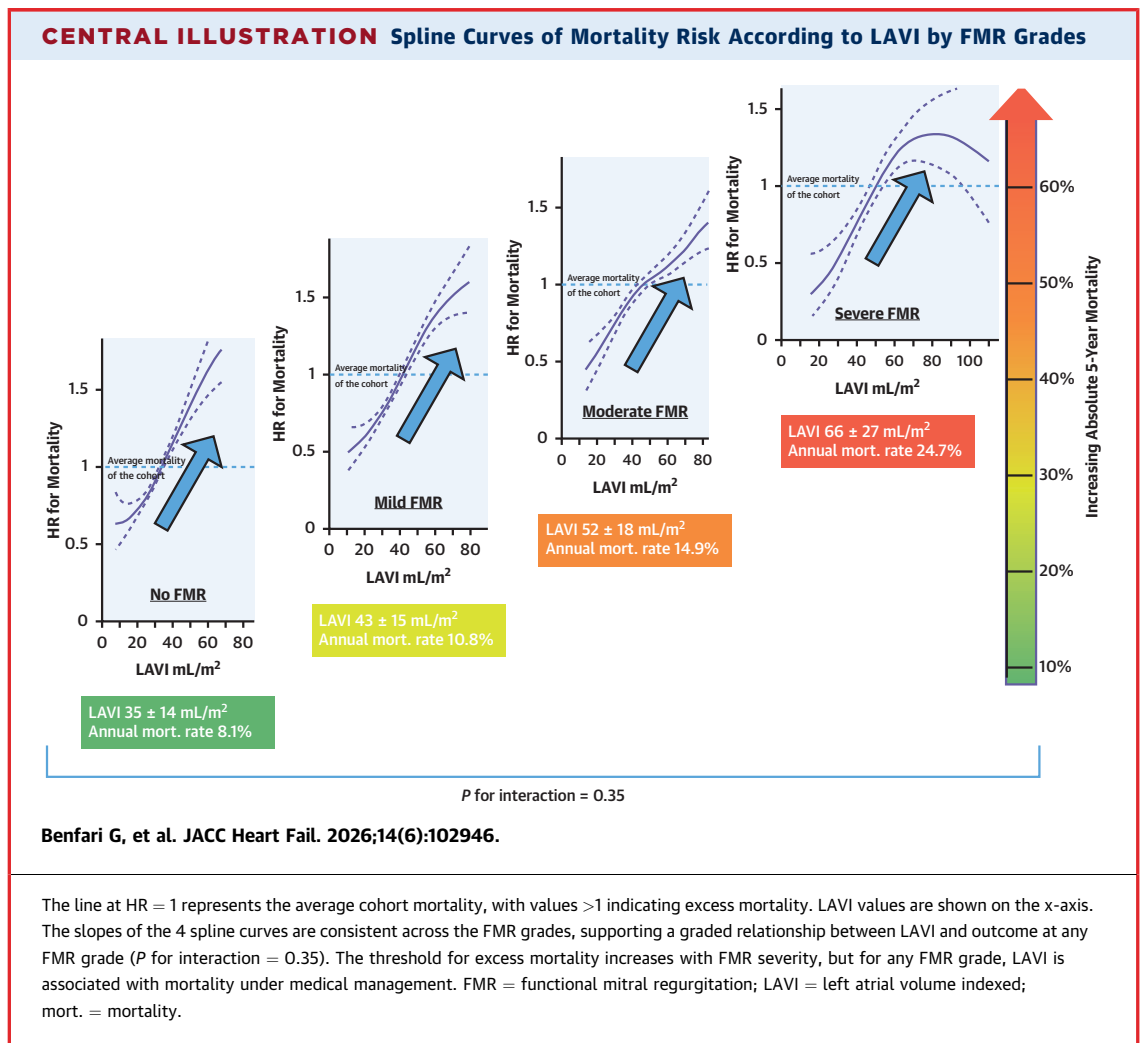


components of prominent prognostic HF scores.⁸ With its considerable power, our study provides a clear answer regarding this uncertainty, showing that all these determinants explain only approximately one-third of LAVI variability, highlighting the role of large individual variability in affecting survival after diagnosis.

LA REMODELING INDEPENDENTLY LINKED TO SURVIVAL. Rationales for evaluating LA remodeling in HFrEF include the need to improve HF prognosis determination,³¹ with new therapies and interventions, such as mitral³² or tricuspid³³ valve repair or ventricular assist device implantation,³⁴ becoming essential components of HF management. Up to now, very few objective measures of cardiac function obtained by imaging are included among scores for HF prognostication, apart from EF.⁸ Although the independent role of some determinants of LA enlargement, such as FMR severity,⁷ E/e',⁶ and associates of LA remodeling including pulmonary hypertension,³⁵ has been shown to profoundly affect HFrEF outcome, the independent role of the LA remained in doubt. Data using surrogates for LA volume (ie, LA linear dimensions or areas⁹⁻¹²) were limited and too crude to fill the need for precise prognostication. For LAVI, a positive association with outcome¹³⁻¹⁶ remains balanced by studies that did not find a significant link between LAVI and outcome,^{17,18} leaving doubts that warrant new data

with detailed measurements of covariates, very large size, and long-term follow-up. This knowledge gap is filled by the present study, whereby higher LAVI measured prospectively in routine practice is strongly and incrementally linked to poor survival after diagnosis, adjusted for the major measure of aging and all predictors of survival associated with LA enlargement, as well as right-sided associates of LA remodeling and particularly the MAGGIC score. Thus, it is now clear that individual variability in the LA response to HFrEF demonstrates a powerful independent link to outcome.

The individual variability in LAVI presents a challenge in defining an optimal prognostic cutoff for excess mortality, as thresholds may vary in specific clinical scenarios as the mentioned determinants are represented and/or combined differently.³⁶ Our spline curve analysis clearly shows that a general threshold for excess risk appears around 40 mL/m², similarly to findings in other conditions.³⁷ The validity of this was further explored through a sensitivity analysis using a Forest plot, demonstrating that in each subgroup this threshold is associated with excess risk. The second threshold of HR crossing 1.5 is 60 mL/m², which also corresponds to thresholds found in other diseases and is valid across all subsets in the Forest plot. Thus, although LA remodeling has historically been considered a secondary response



to valvular or ventricular abnormalities, the present evidence suggests that LAVI is directly linked to excess mortality risk in HFrEF,^{6,38} as it is in other conditions.^{37,39,40}

One of the issues that remained unclear is the specific association of LA remodeling with prognosis in patients with FMR,^{7,41,42} as FMR itself directly causes LA enlargement. FMR may induce maladaptive myocardial changes, including fibrotic replacement and myocyte apoptosis, leading to reduced atrial compliance, increased pulmonary pressure, higher natriuretic peptide activation, and worse functional capacity and outcome.⁴³⁻⁴⁵ LA remodeling has even been suggested as a marker of response to interventional treatment,^{38,46,47} although this issue remains in doubt and is not the subject of our study. Thus, what our data demonstrate clearly is that despite increasing LAVI and increasing mortality rates associated with higher grades of

FMR, the relationship between any increment in LAVI and incremental mortality remains stable, with similar spline curve slopes, throughout the spectrum of FMR severity.

Thus, in all subsets of HFrEF, with all adjustments, LAVI measured in routine practice is an independent and strong predictor of survival after the diagnosis, so its measurement should be encouraged in all laboratories routinely and its inclusion in prognostic scores should be proposed to the cardiology community.

STUDY STRENGTHS AND LIMITATIONS. First, our cohort was not prospectively enrolled with protocolized care but identified retrospectively in a single referral center. However, all measurements were performed prospectively at the time of first diagnosis, unaltered for analysis, and patients were treated by their personal physicians on the basis of these values. Hence, our cohorts are highly

representative of routine clinical practice, particularly for the multiplicity of sonographers and cardiologists involved in the evaluation; this aspect, combined with the robust link between LAVI and outcome, can ultimately be seen as a strength. It is also not a clinical trial and cannot report the effectiveness of therapies. Second, no repeated measurements of LA volume were recorded over time; multiple echocardiograms may have provided further insight into the link between LAVI and outcome in HF. Third, LAVI size was not verified by a core laboratory and LA function analysis was not available for the study cohort. However, such verification, useful to harmonize clinical trials, is not applicable to evaluation in routine practice. Fourth, the present study is not population based, but rather from a single tertiary referral center. Therefore, it reflects routine practice, but the similarity of patients with moderate-severe FMR in our cohort to patients with the same diagnosis in the community⁴⁸ and to patients enrolled in the medical arm of COAPT (Cardiovascular Outcomes Assessment of the MitraClip Percutaneous Therapy for Heart Failure Patients with Functional Mitral Regurgitation Trial) is quite reassuring.³²

Lastly, the medication dose was not analyzed as a covariate, and the levels of mineralocorticoid receptor antagonists were lower than the guideline-recommended levels. Although this is common in observational studies during the years of enrollment, we cannot rule out that contemporary medical therapy may have influenced both LAVI size and clinical outcomes.

CONCLUSIONS

LAVI independently predicts mortality in newly diagnosed stage B or C heart failure, with excess risk

from ≥ 40 mL/m². Its prognostic value shows a graded relationship across all FMR severity grades, highlighting the role of left atrial remodeling in risk stratification specifically among patients with FMR.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: This study demonstrates that LAVI is a strong independent predictor of long-term mortality in patients with heart failure with reduced ejection fraction, even after adjusting for the severity of left ventricular dysfunction, functional mitral regurgitation, atrial fibrillation, and pulmonary hypertension. Moreover, it explores the link between LAVI and outcome across FMR grades. Routine measurement and reporting of LAVI can enhance risk stratification and help identify patients who may benefit from earlier intervention or closer surveillance.

TRANSLATIONAL OUTLOOK: Although LAVI is easily obtained from standard echocardiography, its clinical role had not been clarified at all FMR severity grades. Future studies should determine whether actions capable of reducing LAVI, such as aggressive medical therapy, rhythm control strategies, or mitral valve interventions, always translate into improved clinical outcomes.

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KEY WORDS echocardiography, functional mitral regurgitation, heart failure outcome, left atrial volume, secondary mitral regurgitation

APPENDIX For a supplemental table and figure, please see the online version of this paper.